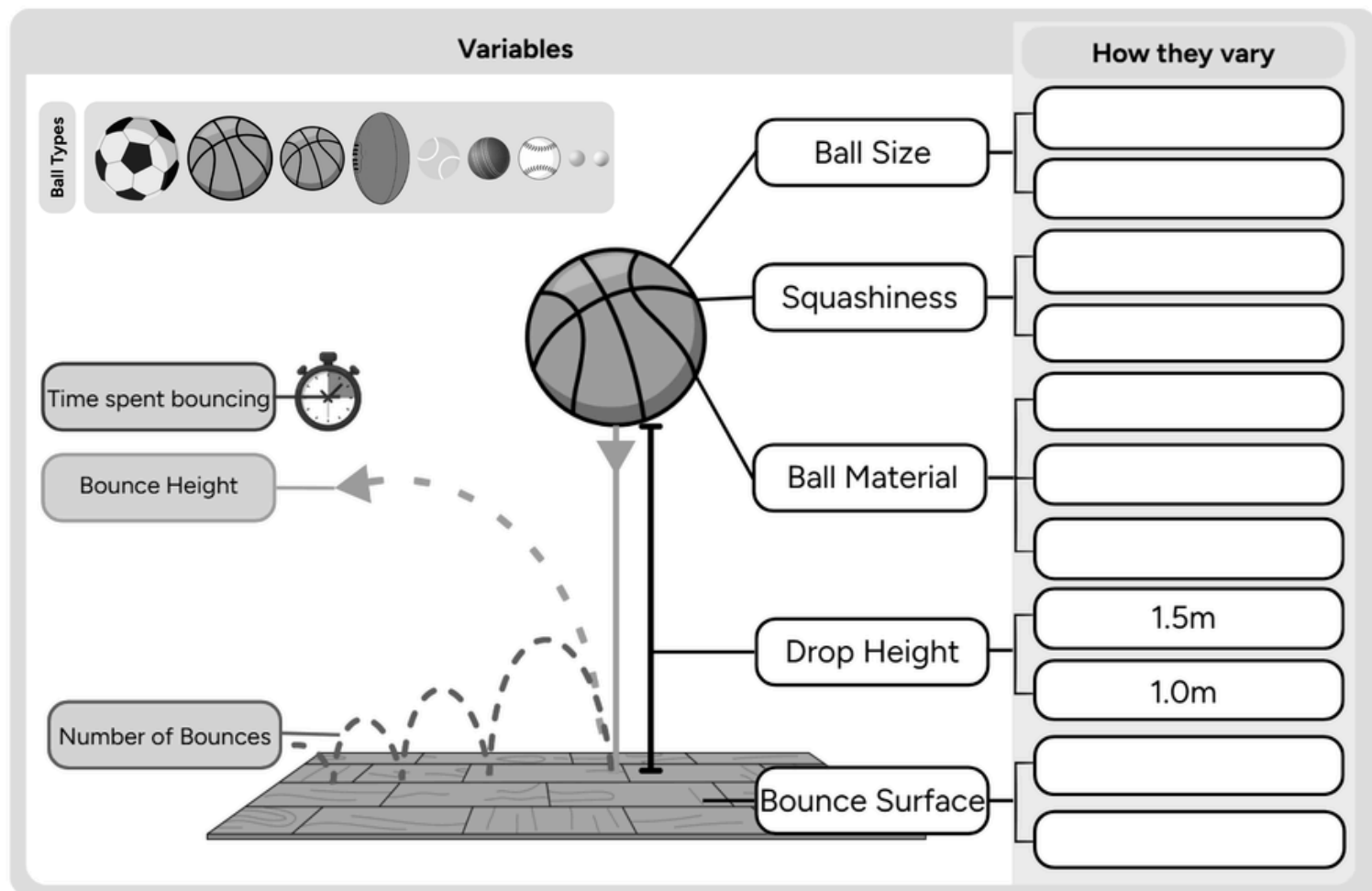


What are variables?

Name: _____

Complete the mindmap to record variables in the features of the balls, in how they are dropped and how they bounce.

When designing a fair test it is good practice to use images and tables to record all of the variables that could have an effect on the result.



Write all the possible variables from the ball bounce scenario above in the table below, the first one has been completed for you:

Time spent bouncing (minutes/seconds)			

Define the different types of variables in a fair test:

Independent variable: _____

Dependent variable: _____

Controlled variables: _____

Plan your own fair test experiment

1 Choosing from the variables you identified on the previous page and complete up to three testable questions about the ball bounce experiment.

Example:

What will changing the ball material do to the number of bounces?

(What are you changing?) (What are you measuring?)

What will changing the _____ do to the _____?

(What are you changing?) (What are you measuring?)

How does changing the _____ affect the _____?

If we change the _____ how does it affect the _____?

Now **highlight** your favourite question and build your reasoned prediction:

2 What am I changing?

Use the following options:

- increase / decrease
- with / without
- condition 1 versus condition 2

3 What am I measuring?

Use the following options:

- increase / decrease
- stay the same

4 Prediction

If I do this... then this will happen.



5 Reason

This is because...



Reasoned prediction

Put it all together - prediction + reason in two sentences.

Draw your fair test experiment as a diagram

Design your experiment based on your testable question. Draw your experiment set up in the space provided, including the tools you will use to measure. Use arrows to label all variables.

Independent Variable

(What are you changing and how?)

Draw your fair test experiment below

Dependent Variable

(What are you measuring and how?)

Controlled Variables

(What are you keeping the same and how?)

You have designed your experiment, now explain why your experiment is an example of a fair test.
